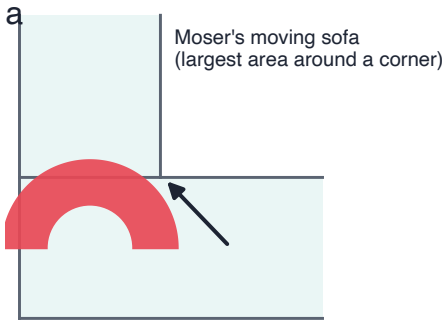
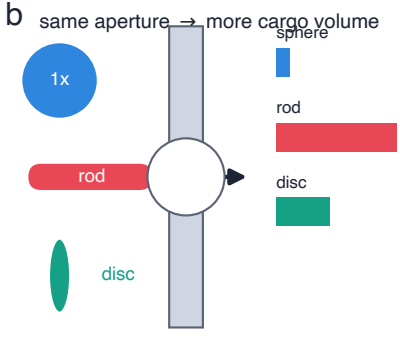
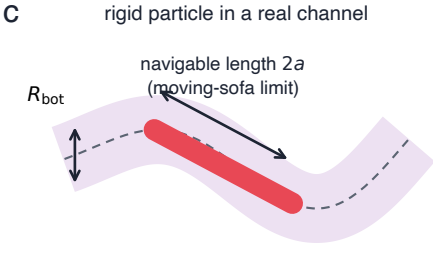
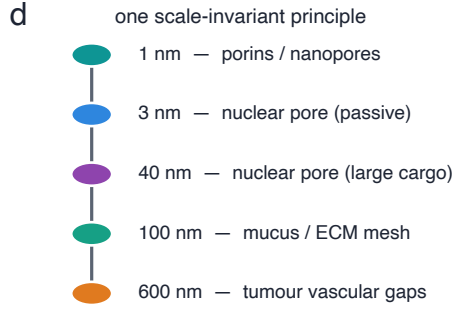

a
Moser's moving sofa
(largest area around a corner)


b
same aperture → more cargo volume
1x
rod
disc
sphere
rod
disc


c
rigid particle in a real channel
navigable length 2a
(moving-sofa limit)
R_{bot}


d
one scale-invariant principle
1 nm — porins / nanopores
3 nm — nuclear pore (passive)
40 nm — nuclear pore (large cargo)
100 nm — mucus / ECM mesh
600 nm — tumour vascular gaps